

Euclid's Elements — Book II

Proposition 14

Formal Reconstruction — Technical Documentation

Jurek Rózański

Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJTEAMLAB

To construct a square equal to a given rectilinear figure.

1 Introduction

Proposition II.14 completes the quadrature of rectilinear figures in Book II. Its geometric task is constructive: from an arbitrary rectilinear figure, construct a square equal to it in content [?].

Euclid's proof has two visibly different layers. First, Proposition I.45 replaces the given rectilinear figure by an equal rectangular parallelogram. Second, the rectangle is converted into a square. In the unequal-side case this second layer uses the semicircle construction, Proposition II.5, the Pythagorean theorem I.47, and Common Notion 3.

The production Lean theorem preserves this outer architecture. The input figure is represented by a nonempty `HilbertTriangulatedFigure` whose triangles are nondegenerate. The conclusion is not equality of real-valued areas. It is the synthetic equal-content statement

$$\text{rectilinearTerm}(L) \simeq_{\text{ec}} S_Q,$$

where S_Q is the concrete parallelogram term of a square and $\simeq_{\text{ec}}$ denotes `HilbertScissorsEquicomplementable`.

The production file begins with

```
import CGJteamLab.Proposition2_13
import CGJteamLab.Proposition45
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.Proposition2_5
import CGJteamLab.HilbertRectangle
import CGJteamLab.HilbertSquareTransport
```

The import of `Proposition2_13` maintains the sequential Book II module architecture. The theorem `euclid_proposition_2_13` is not called. Among earlier Book II propositions, the direct mathematical dependency of II.14 is II.5.

The reconstruction remains entirely synthetic. It uses no coordinates, no numerical area function, no segment multiplication, no theory of proportion, no similar triangles, and no trigonometry.

2 Euclid's Statement and Classical Configuration

2.1 The geometric statement

Euclid's statement is:

To construct a square equal to a given rectilinear figure.

Let the given rectilinear figure be A . I.45 first constructs a rectangular parallelogram BD equal to A . In the standard lettering its consecutive vertices may be read as B, C, D, E , with adjacent sides BE and ED .

If

$$BE = ED,$$

the rectangle is already a square. Otherwise one side is greater. Euclid chooses the orientation in which $BE > ED$, produces BE to F , constructs

$$EF = ED,$$

and bisects BF at G . He then describes the semicircle BHF with center G , produces DE to meet it at H , and joins GH .

2.2 The classical order data

The diagram contains the order

$$B - G - E - F$$

and, after the intersection point has been chosen,

$$D - E - H.$$

These order relations are mathematically significant. They determine the internal cut required by II.5 and the right triangle required by I.47.

Euclid's phrase "let BE be greater" is harmless on paper because the rectangle can be read with its two adjacent sides interchanged. Lean cannot leave this symmetry implicit. The production proof performs a segment trichotomy and treats all three cases explicitly.

3 Classical Proof

Assume first that $BE > ED$ and use the construction above. Because G is the midpoint of BF and E is the unequal cut point, II.5 gives

$$\text{Rect}(BE, EF) + \text{Sq}(EG) = \text{Sq}(GF).$$

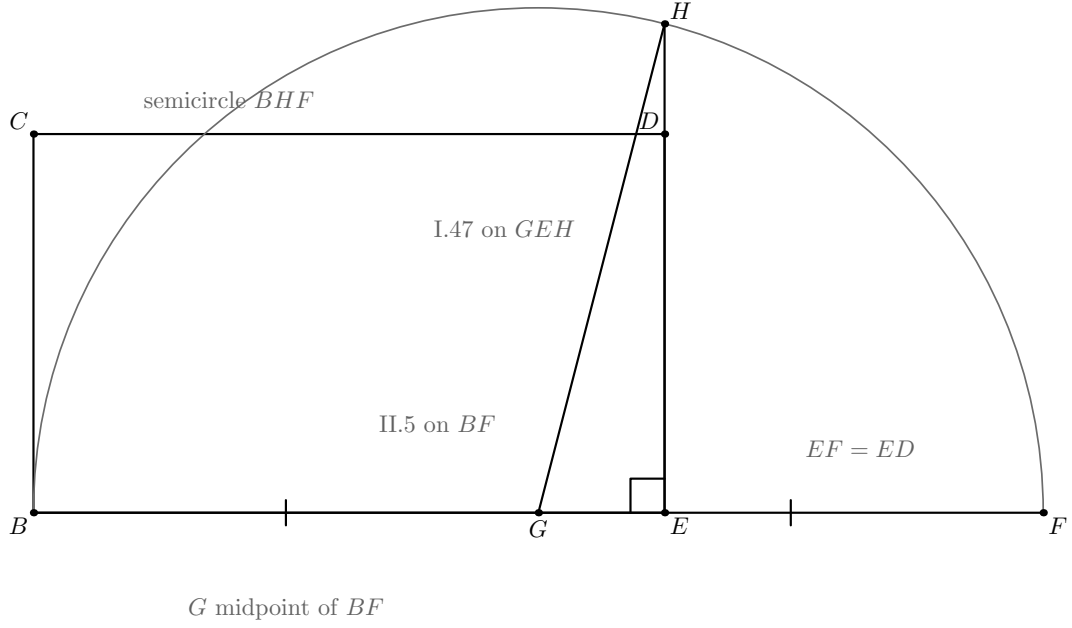


Figure 1: The classical unequal-side configuration. The rectangle $BCDE$ represents Euclid's rectangular parallelogram BD . The side BE is produced to F with $EF = ED$, G is the midpoint of BF , and the semicircle centered at G meets the produced line DE at H . Proposition II.5 is applied on the baseline BF , while I.47 is applied to the right triangle GEH .

Since H lies on the semicircle centered at G ,

$$GF = GH.$$

Hence

$$\text{Rect}(BE, EF) + \text{Sq}(EG) = \text{Sq}(GH).$$

The triangle GEH is right at E , so I.47 gives

$$\text{Sq}(GH) = \text{Sq}(GE) + \text{Sq}(EH).$$

Combining the two equalities,

$$\text{Rect}(BE, EF) + \text{Sq}(EG) = \text{Sq}(EG) + \text{Sq}(EH).$$

Common Notion 3 removes the common square on EG :

$$\text{Rect}(BE, EF) = \text{Sq}(EH).$$

But $EF = ED$, so the rectangle contained by BE, EF equals the original rectangle BD . Since BD was constructed by I.45 equal to the original rectilinear figure A , the square on EH is equal to A .

The local classical dependency graph is therefore

$$\begin{array}{c}
\text{given rectilinear figure} \\
\Downarrow \\
\text{I.45: equal rectangular parallelogram } BD \\
\Downarrow \\
BE = ED \quad \text{or} \quad BE \neq ED \\
\Downarrow \\
\text{I.3} + \text{I.10: } EF = ED, \text{ } G \text{ midpoint of } BF \\
\Downarrow \\
\text{semicircle } BHF, \quad D - E - H \\
\Downarrow \\
\text{II.5: } R_{BE,EF} + S_{EG} = S_{GF} \\
\Downarrow \\
GF = GH, \quad \text{I.47 on } GEH \\
\Downarrow \\
R_{BE,EF} + S_{EG} = S_{EH} + S_{EG} \\
\Downarrow \\
\text{Common Notion 3} \\
\Downarrow \\
R_{BE,EF} = S_{EH} \\
\Downarrow \\
EF = ED, \quad BD = A \\
\Downarrow \\
\text{II.14.}
\end{array}$$

Only II.5 is used from Book II. In particular, the proof does not require proportions or similar triangles. Those give a later interpretation of EH as a mean proportional, but they are not part of Euclid's Book II proof and are not part of the Lean reconstruction.

4 The Geometric Content Identity

The formal proof translates the central classical calculation into the scissors/equicomplementability layer.

For concrete representatives it first obtains

$$R_{EB,FE} + S_{EG} \simeq_{ec} S_{GF}.$$

The II.5 core is actually stronger: before representative transport it is an exact `HilbertScissorsEq`. It is weakened to `HilbertScissorsEquicomplementable` when the independently constructed square on GB is transported to a square on GF .

The Pythagorean block gives

$$S_{GF} \simeq_{ec} S_{EG} + S_{EH}.$$

After transitivity and a formal commutation of the right-hand sum,

$$R_{EB,FE} + S_{EG} \simeq_{ec} S_{EH} + S_{EG}.$$

The private cancellation theorem

```
proposition2_14_equicomplementable_cancel_right
```

implements Common Notion 3 and yields

$$R_{EB,FE} \simeq_{\text{ec}} S_{EH}.$$

This is not subtraction in an additive group. The proof unfolds the definition of equicomplementability, enlarges the common complements by the same term, and reassociates the finite formal sums. No negative scissors term and no numerical area difference is introduced.

Finally $EF \cong ED$ and rectangle transport identify the constructed rectangle representative with the original rectangle. Thus

$$R_{BCDE} \simeq_{\text{ec}} S_{EH}.$$

5 The Semicircle Intersection in the Formal Reconstruction

5.1 Why the construction is reorganized

The current geometry API does not provide a convenient theorem that directly constructs the intersection of the produced line DE with a circle centered at G while simultaneously returning all order data needed later. Rather than postulating such a specialized operation, II.14 reconstructs the same point H from the available synthetic circle-continuity interface.

The key auxiliary point is K , chosen so that E is the midpoint of GK . Two circles are considered:

$$\mathcal{C}(G, GF), \quad \mathcal{C}(K, GF).$$

Their centers are symmetric about E and they have equal radii. The theorem `hilbert_circle_circle_intersection` supplies a common point H .

Because

$$GH = KH$$

and E is the midpoint of GK , synthetic perpendicular-bisector geometry forces H onto the line through E perpendicular to GK . The rectangle provides exactly such a perpendicular carrier through D, E . The proof then chooses the orientation with

$$D - E - H.$$

Thus the two-circle construction is not an alternative content proof. It is an auxiliary construction of the classical intersection point required by the same II.5–I.47 argument.

5.2 Continuity cost

This is the only place where the final axiom audit visibly records the circle-continuity principle

```
Geometry.hilbert_circle_circle_intersection
```

The dependency is mathematically meaningful: a circle intersection is actually constructed. It is an inherited foundational principle, not a new axiom introduced by Proposition II.14.

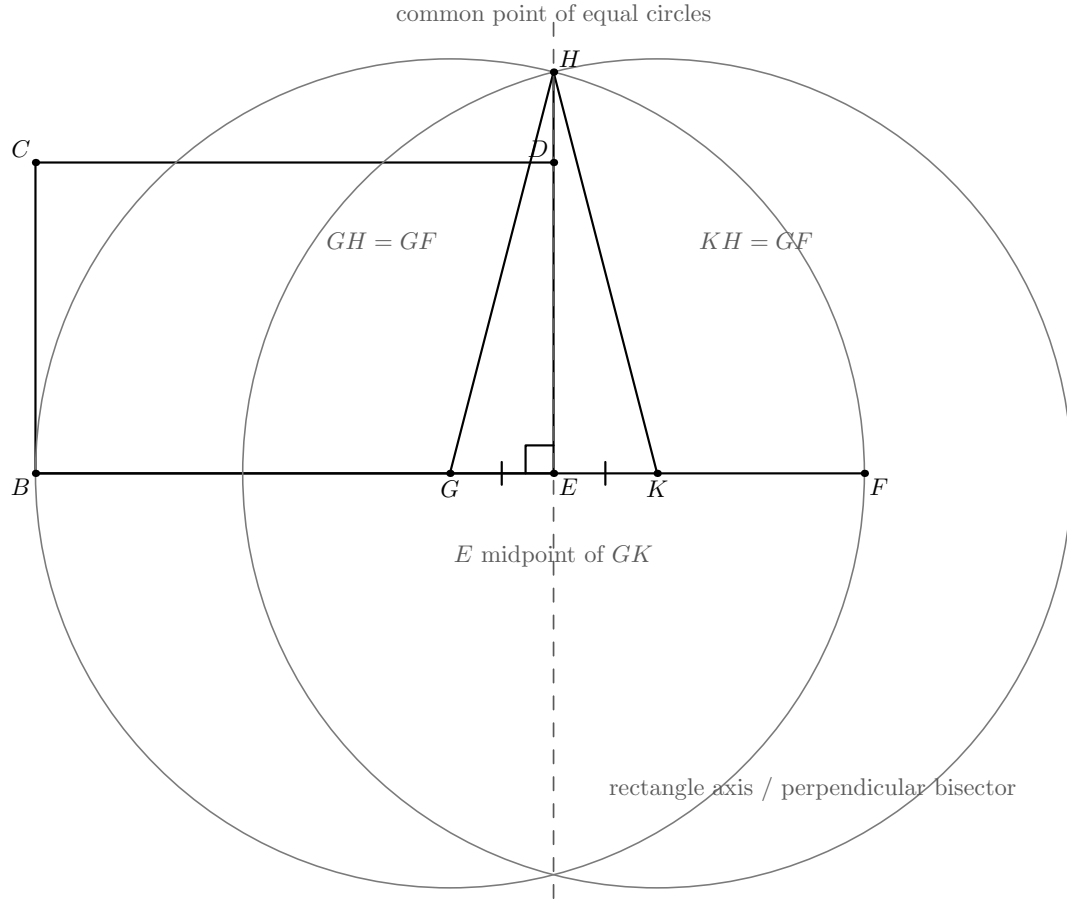


Figure 2: The formal replacement for the direct line–semicircle intersection. The point E is the midpoint of GK , and the equal circles centered at G and K have radius congruent to GF . Their upper intersection H is equidistant from G and K , hence lies on the perpendicular bisector through E . The rectangle identifies this perpendicular with the carrier DE ; the point is then oriented so that $D - E - H$. In this drawing the resulting H is the same geometric point as in Figure 1.

6 Reusable Book II Infrastructure

6.1 Proposition II.5

The direct earlier Book II theorem is

```
euclid_proposition_2_5
```

The proposition-local adapter first constructs the full geometric witness package demanded by the configured II.5 API. The helper

```
proposition2_14_rectangle_cut_exists
```

constructs the required rectangle cut, diagonal intersection, and parallelogram pieces.

The theorem

```
proposition2_14_ii5_exists
```

then packages the II.5 conclusion with concrete representatives of

$$R_{EB,FE}, \quad S_{EG}, \quad S_{GB}.$$

The next normalization,

```
proposition2_14_ii5_normalized_GF
```

uses midpoint congruence and `hilbert_square_transport` to replace the square on GB by a square on GF .

6.2 Rectangle representation and transport

The proof uses the general `HilbertRectangle` API rather than a primitive multiplication of segment lengths. In particular:

- `IsRectangleContainedBy` records a concrete rectangle representative and the two congruent containing sides;
- `rectangle_contained_by_swap` cyclically rotates a rectangle while supplying an exact scissors equality between the two presentations;
- `rectangle_transport_scissorsEq` transports a rectangle between congruent containing-side pairs.

These operations explain both the final substitution $EF \cong ED$ and the formal handling of Euclid’s phrase “let BE be greater.”

6.3 Square transport

I.46 and I.47 are existential at the level of square vertices. The proof therefore cannot assume that two squares on congruent segments are literally the same four points. The proposition-independent theorem

```
hilbert_square_transport
```

converts congruence of the bases into equicomplementability of the concrete square representatives.

7 Proof Architecture of the Reconstruction

The production proof separates naturally into four blocks.

7.1 Block I: rectangle side trichotomy

For a rectangle $BCDE$, the adjacent side magnitudes BE and ED satisfy segment trichotomy.

- (1) If $BE \cong ED$, a cyclic presentation of the rectangle has equal adjacent sides and is already a square.
- (2) If $BE < ED$, the rectangle is rotated once and the unequal-side construction is applied in the orientation where the longer side plays Euclid's role of BE .
- (3) If $ED < BE$, the unequal-side construction is applied directly.

The theorem

```
proposition2_14_rectangle_to_square
```

therefore proves the rectangle-level statement without assuming beforehand which side is longer.

7.2 Block II: unequal-side construction and the point H

The helper

```
proposition2_14_unequal_side_order
```

uses segment extension, segment construction, and midpoint existence to build the classical baseline data:

$$B - G - E - F, \quad EF \cong ED, \quad G \text{ midpoint of } BF.$$

The next circle-geometry chain constructs K with E the midpoint of GK , establishes inside/outside witnesses for the two equal circles, invokes circle-circle continuity, proves that the common point lies on the rectangle axis, and orients it so that

$$D - E - H, \quad GH \cong GF, \quad KH \cong GF.$$

The packaged endpoint is

```
proposition2_14_oriented_circle_point
```

7.3 Block III: II.5, I.47, and Common Notion 3

The II.5 package gives

$$R_{EB,FE} + S_{EG} \simeq_{\text{ec}} S_{GF}.$$

The theorem

```
proposition2_14_pythagorean_GF
```


proves

$$S_{GF} \simeq_{ec} S_{EG} + S_{EH}$$

by I.47 on the right triangle GEH followed by square transport.

Then

```
proposition2_14_cancel_EG
```

performs the Common-Notion 3 cancellation and

```
proposition2_14_original_rectangle_equals_square
```

transports the rectangle on EB, EF back to the original rectangle.

7.4 Block IV: from an arbitrary rectilinear figure to a rectangle

The public theorem begins with a nonempty triangulated figure L . Its first nondegenerate triangle supplies a nonzero side. I.46 constructs a square on that side, and one corner of the square supplies a nondegenerate right angle.

I.45 is then invoked with this right angle as the prescribed angle. It constructs a parallelogram equicomplementable with the whole rectilinear figure. Since its prescribed angle is congruent to a right angle, right-angle transport proves that this parallelogram is an **IsRectangle**.

The rectangle-level theorem from Block I produces the square, and transitivity with the I.45 content relation gives the public II.14 conclusion.

This auxiliary use of I.46 is a formal interface device. It avoids adding an extra right-angle parameter to the public theorem.

8 Lean Formalization

8.1 The proposition-local theorem chain

The production module is large because it makes the semicircle-intersection and configured II.5 witness geometry explicit. Its declarations are grouped conceptually as follows.

Rectangle and order setup.

```
proposition2_14_rectangle_equal_sides_square  
proposition2_14_unequal_side_order  
proposition2_14_inside_circle  
proposition2_14_right_triangle_after_H  
proposition2_14_mirror_center
```

Perpendicular-bisector and circle geometry.

```
proposition2_14_rectangle_axis_equidistant  
proposition2_14_equidistant_right_angle
```

```

proposition2_14_same_side_angle_collinear
proposition2_14_opposite_side_reduction
proposition2_14_right_angle_axis_collinear
proposition2_14_centers_less_two_radii
proposition2_14_circle_outside_witness
proposition2_14_less_cancel_right
proposition2_14_circle_inside_witness
proposition2_14_equal_circles_intersect
proposition2_14_circle_point_on_rectangle_axis
proposition2_14_circle_point_off_center_line
proposition2_14_circle_point_on_DE
proposition2_14_oriented_copy_on_extension
proposition2_14_oriented_point_stays_on_circles
proposition2_14_oriented_circle_point

```

II.5 witness construction and content normalization.

```

proposition2_14_ii5_baseline
proposition2_14_rectangle_cut_core_aux      -- private
proposition2_14_rectangle_cut_between      -- private
proposition2_14_diagonal_cut_intersection  -- private
proposition2_14_rectangle_cut_exists
proposition2_14_ii5_exists
proposition2_14_ii5_normalized_GF

```

Pythagoras, cancellation, and final wrappers.

```

proposition2_14_pythagorean_GF
proposition2_14_equicomplementable_cancel_right -- private
proposition2_14_cancel_EG
proposition2_14_original_rectangle_equals_square
proposition2_14_rectangle_to_square
euclid_proposition_2_14

```

The source order reflects the incremental implementation. The mathematical DAG is much shorter:

$$\begin{array}{c}
\text{rectangle side trichotomy} \\
\Downarrow \\
\text{unequal-side baseline + synthetic circle intersection} \\
\Downarrow \\
\text{II.5 + I.47 + Common Notion 3} \\
\Downarrow \\
\text{rectangle} \simeq_{ec} \text{square} \\
\Downarrow \\
\text{I.45 outer reduction} \\
\Downarrow \\
\text{rectilinear figure} \simeq_{ec} \text{square.}
\end{array}$$

8.2 The public theorem

The following is an ASCII transcription of the current compiling public signature:

```

theorem euclid_proposition_2_14
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (L : HilbertTriangulatedFigure Geo)
  (hNonempty : Ne L [])
  (hTriangles :
    forall t, List.Mem t L ->
      Not (Collinear Geo t.A t.B t.C)) :
  exists Q0 Q1 Q2 Q3 : Geo.Point,
    IsSquare Geo Q0 Q1 Q2 Q3 /\
    HilbertScissorsEquicomplementable Geo
      (rectilinearTerm Geo L)
      (hilbertParallelogramTerm Geo Q0 Q1 Q2 Q3) := by

```

The theorem exposes only the four vertices of the final square, its `IsSquare` certificate, and the final equal-content relation. It does not expose the I.45 parallelogram, the selected right-angle square, the points F, G, K, H , the II.5 rectangle representative, the intermediate square representatives, or the circle witnesses.

8.3 Representation of the input figure

The input is a `HilbertTriangulatedFigure`, represented at the current interface as a finite list of triangles. The hypotheses state:

- the list is nonempty;
- every triangle in it is noncollinear.

The term `rectilinearTermGeoL` is therefore the formal content representative of the whole rectilinear figure. II.14 does not assume that the input is already a rectangle or a single polygon with a distinguished boundary cycle.

9 Relationship with Earlier Book II Propositions

9.1 The classical dependency is II.5

Euclid's only earlier Book II proposition in the local proof is II.5. The production source calls

```
euclid_proposition_2_5
```

inside the proposition-local existential adapter.

The use is exact in mathematical content:

$$R_{BE,EF} + S_{EG} = S_{GF}.$$

The remaining steps are Book I geometry, Common Notion 3, and representative transport.

9.2 The Proposition2_13 import is architectural

The file imports `Proposition2_13`, but neither `euclid_proposition_2_13` nor a proposition-local `II.13` helper is called. This is therefore not a classical dependency edge

$$\text{II.13} \longrightarrow \text{II.14}.$$

The correct distinction is

classical Book II DAG: II.5

Lean module DAG: `Proposition2_13` + `Proposition2_5` + Book I/shared modules.

10 Relationship with Book I Infrastructure

The principal Book I roles are:

- I.45 converts the arbitrary rectilinear figure into an equal parallelogram with a prescribed right angle, hence into a rectangle;
- I.46 constructs the auxiliary square that supplies a right-angle witness for the I.45 call and constructs several target squares in proposition-local blocks;
- I.47 supplies the Pythagorean equal-content relation in triangle GEH .

Classically, I.3 and I.10 are named explicitly for the construction $EF = ED$ and the midpoint G of BF . The Lean proof reaches the same synthetic data through the established segment-construction and midpoint-existence interfaces rather than by calling proposition wrappers solely to reproduce Euclid’s citation labels.

The proof also relies on established Hilbert/Book Zero infrastructure for strict betweenness, segment comparison and trichotomy, congruence transport, right-angle transport, perpendicular-bisector uniqueness, same-side and opposite-side reasoning, circle inside/outside witnesses, parallel geometry, Pasch-style intersection constructions, rectangle representatives, and finite scissors calculus.

11 Formalization Notes and Hidden Geometric Data

11.1 Euclid’s “let BE be greater” becomes a real case split

The classical proof simply selects the longer adjacent side. Lean proves segment trichotomy for BE and ED .

The equality branch immediately produces a square. In the branch $BE < ED$, the rectangle is cyclically rotated so that the longer side enters the unequal-side construction in the required position. In the branch $ED < BE$, the construction is used directly.

This is a genuine formalization obligation hidden by the symmetry of Euclid’s diagram.

11.2 The point H is constructed, not read from a picture

The classical phrase “produce DE to H ” presupposes the relevant line–semicircle intersection. The Lean proof must supply existence, incidence, and orientation.

The equal-circle construction first produces an unoriented common point. Subsequent lemmas prove:

$$H \in DE, \quad H \neq E,$$

and then replace or orient the witness so that the exact strict order

$$D - E - H$$

holds while preserving the two circle congruences.

11.3 The right angle GEH is derived from rectangle geometry

Once $B - G - E - F$ and $D - E - H$ are available, the carriers GE and DE are the same carriers as the two perpendicular sides through E of the original rectangle. The helper

```
proposition2_14_right_triangle_after_H
```

therefore proves that $\angle GEH$ is right. This is the exact hypothesis needed for I.47.

11.4 Common Notion 3 is content cancellation, not arithmetic subtraction

The theorem

```
proposition2_14_equicomplementable_cancel_right
```

is private to the production module. It proves

$$P + T \simeq_{\text{ec}} Q + T \implies P \simeq_{\text{ec}} Q.$$

Its proof stays inside the definition of equicomplementability and finite formal sums. It does not construct an additive inverse $-T$.

This is the formal counterpart of Euclid’s instruction to subtract the same square on GE from both equal contents.

11.5 Representative normalization is unavoidable

II.5, I.46, and I.47 may return different concrete squares or rectangles on congruent segments. The proof therefore distinguishes:

- congruence of base segments;
- equality in content of concrete square representatives;
- exact scissors equality of cyclic rectangle presentations.

The transport lemmas make these identifications explicit rather than treating all squares on a given length as definitionally equal.

11.6 The public I.45 wrapper needs a prescribed angle

The final theorem should depend only on the input rectilinear figure, but I.45 asks for a nondegenerate angle to be copied. The proof extracts the first triangle from the nonempty list, takes one nonzero side, constructs a square on it by I.46, and uses a corner of that square as the prescribed right angle.

This witness disappears from the public conclusion. It is interface glue, not an additional geometric hypothesis.

12 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. The production module compiled cleanly, and the full project `lake build` was reported clean.

The transitive audit

```
#print axioms Geometry.euclid_proposition_2_14
```

reported

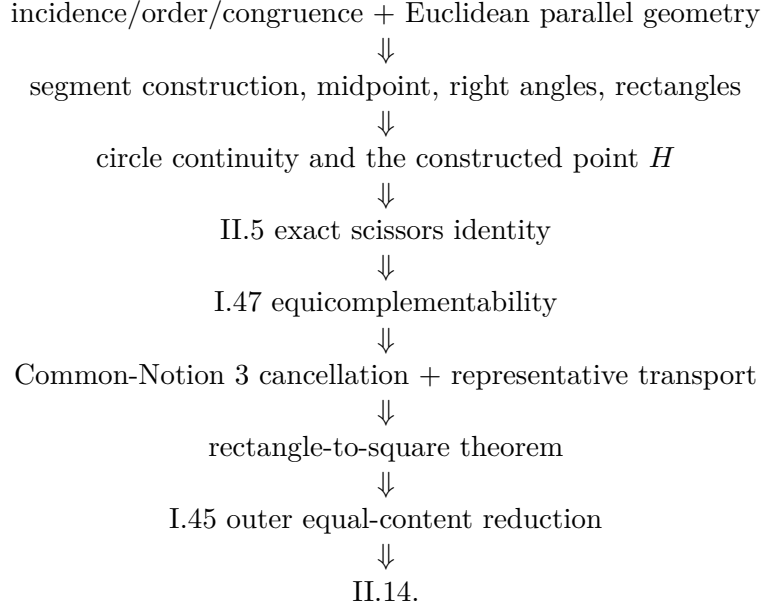
```
[propext,  
 Classical.choice,  
 Geometry.bookZero_nullSegment1,  
 Geometry.bookZero_nullSegment2,  
 Geometry.hilbert_circle_circle_intersection,  
 Quot.sound]
```

These are inherited project dependencies. In particular, `hilbert_circle_circle_intersection` records the genuine continuity step used to construct the common point of the two circles. The appearance of `bookZero_nullSegment2`, which was not present in the reported II.13 audit, is also transitive: II.14 does not declare a new null-segment axiom.

The precise status is therefore:

Proposition II.14 adds no new axiom and no `sorry`; it is not literally axiom-free.

The semantic layers are



No numerical area function, segment field, segment multiplication, coordinate model, similar-triangle theory, proportion theory, trigonometric function, or numerical angle measure is used.

13 Source Interpretation

Hartshorne’s equal-content viewpoint is the appropriate semantic control for the phrase “equal to a given rectilinear figure.” It keeps the Book II construction in a geometric content layer rather than silently replacing it by equality of real-valued areas [?].

Hilbert supplies the broader architectural separation between incidence/order/congruence, equidecomposition and equicomplementability, and the later arithmetic of segments [?]. Proposition II.14 is especially instructive because its modern mnemonic

$$(\text{rectangle with sides } a, b) \longmapsto (\text{square with side } \sqrt{ab})$$

must not be read back into the Lean proof as multiplication or square roots. The formal proof never enters that arithmetic layer.

Greenberg remains a foundational and semantic cross-check for Hilbert-style order, congruence, perpendicular geometry, and continuity, and for the distinction between geometric content and numerical area [?]. The project’s particular `HilbertScissorsEquicomplementable` implementation is project infrastructure rather than a theorem attributed to Greenberg.

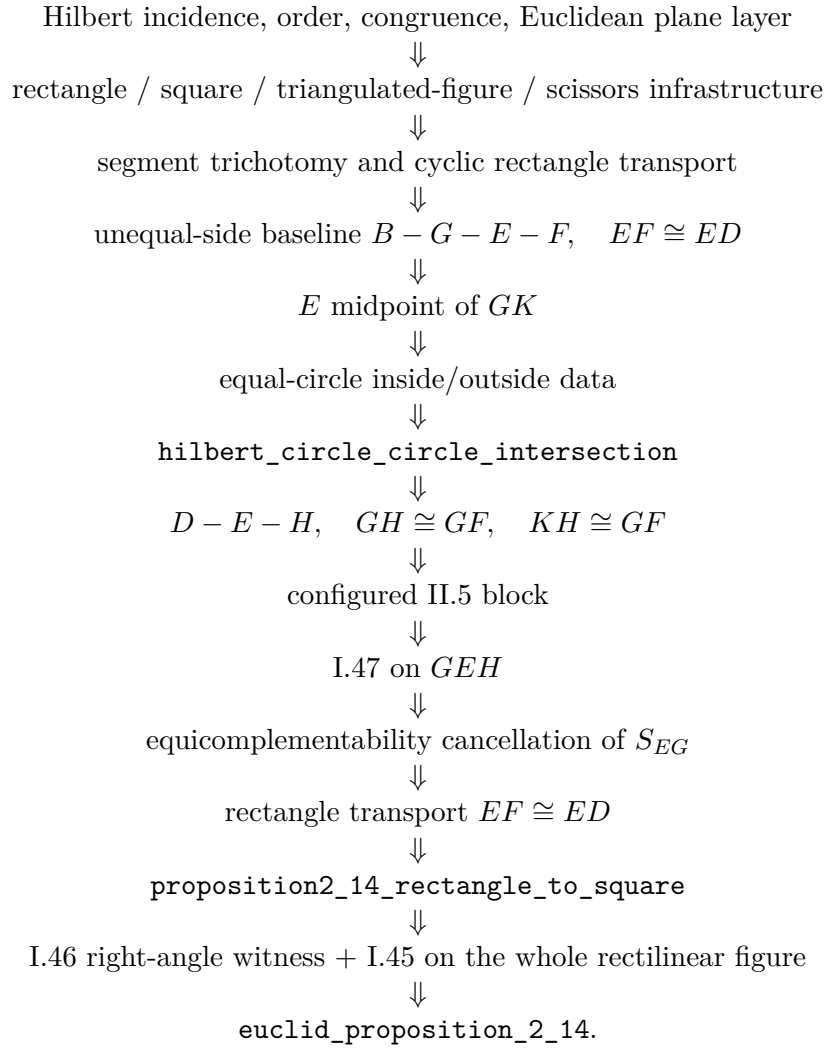
Borsuk–Szmielw are relevant to the incidence/order/congruence and representation shell: midpoint constructions, perpendicular and parallel geometry, positional reasoning, and the finite representation of rectilinear figures belong on this side of the architecture [?]. The current reconstruction does not attribute its complete scissors calculus to their text.

14 Dependency Summary

The classical local chain is

$$\boxed{\text{I.45} \longrightarrow \text{I.3} + \text{I.10} \longrightarrow \text{semicircle} \longrightarrow \text{II.5} \longrightarrow \text{I.47} \longrightarrow \text{CN3} \longrightarrow \text{II.14.}}$$

The actual production chain is



The implementation checkpoint is:

New geometric axiom in II.14:	no
New content axiom in II.14:	no
New proposition-local helper theorem:	yes
New reusable helper promoted to a shared module:	no
New <code>HilbertRectangle</code> theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.5
Imports II.13 module:	yes, architectural only
Calls <code>euclid_proposition_2_13</code> :	no
Uses I.45:	yes
Uses I.46:	yes
Uses I.47:	yes
Uses circle continuity:	yes, inherited
Uses exact <code>HilbertScissorsEq</code> internally:	yes
Public conclusion is exact <code>HilbertScissorsEq</code> :	no
Public conclusion is equicomplementability:	yes
Uses content cancellation:	yes, CN3 form
Uses numerical area:	no
Uses segment multiplication:	no
Uses proportions / similar triangles:	no
Production source compile:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local <code>sorry/admit</code> :	no
Transitive <code>#print axioms</code> :	six inherited dependencies; see audit above

15 Conclusion

Proposition II.14 is the endpoint of Book II’s quadrature program for rectilinear figures. Its classical proof is short:

$$\text{I.45} \rightarrow \text{II.5} \rightarrow \text{I.47} \rightarrow \text{Common Notion 3.}$$

The formal reconstruction preserves that content architecture while making three diagrammatic assumptions explicit.

First, Euclid’s choice of the longer rectangle side becomes a complete segment-trichotomy argument with cyclic rectangle transport. Second, the line–semicircle intersection is reconstructed through two equal circles and the project’s circle–circle continuity principle; the resulting point is then proved to lie on the correct perpendicular carrier and is oriented as $D - E - H$. Third, Common Notion 3 is implemented as cancellation inside equicomplementability rather than as numerical subtraction.

The final public theorem states exactly the intended synthetic quadrature: for every nonempty triangulated rectilinear figure with nondegenerate triangles, there exists a concrete square equicomplementable with the whole figure,

$$\boxed{\text{rectilinearTerm}(L) \simeq_{ec} S_Q.}$$

No area number, product of segment lengths, square root, proportion, similar triangle, coordinate system, or trigonometric identity is required.